

City: Recife, Brazil

Phosphorus from Africa reaches Recife, Brazil, primarily through massive dust plumes carried by trade winds across the Atlantic, originating from sources like the Bodélé Depression in Chad (rich in fish bone/scale remains) and Saharan soils, acting as vital fertilizer for the Amazon, with some settling in the ocean and coastal areas like Recife's Atlantic waters, replenishing nutrients.

Here's [how the process works](#):

1. Source Material: Huge amounts of dust, rich in phosphorus (apatite), are lifted from dry African landscapes, particularly the Sahara and the ancient lakebed of Mega-Lake Chad (Bodélé Depression).
2. Wind Transport: Strong winds, including the trade winds, pick up these fine particles and carry them thousands of miles across the Atlantic Ocean towards South America.
3. Atmospheric Delivery: This dust forms vast plumes that travel west, delivering essential phosphorus to the Amazon basin and the surrounding Atlantic Ocean.
4. Deposition in Recife: As these dust clouds move over the tropical Atlantic, particles settle out through rain (wet deposition) and natural settling (dry deposition) into the ocean waters near Brazil's coast, including the region of Recife.
5. Nutrient Replenishment: This natural atmospheric transport system is crucial for fertilizing the Amazon and the North Atlantic, counteracting nutrient loss from heavy rainfall and biological activity, with coastal areas like Recife receiving a share of these airborne nutrients.

So, it's a natural atmospheric conveyor belt moving dust and nutrients from Africa to South America's doorstep.

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